LECTURE - 5 and 6

Self Attention Vs Cross Attention

Self Attention → $\underbrace{\text{query / Key / Value}}_{\text{Same input sequence}}$

e.g. The dog Sat on the sofa

Key points:

→ All positions/words attend to all the other words.

→ Every token will be aware of the entire input

→ BERT/GPT

Cross-Attention

query → 1 source

Key/Value → Another source

Machine translation model:

decoder → English words

Cross attention → encoded

German ↓ English

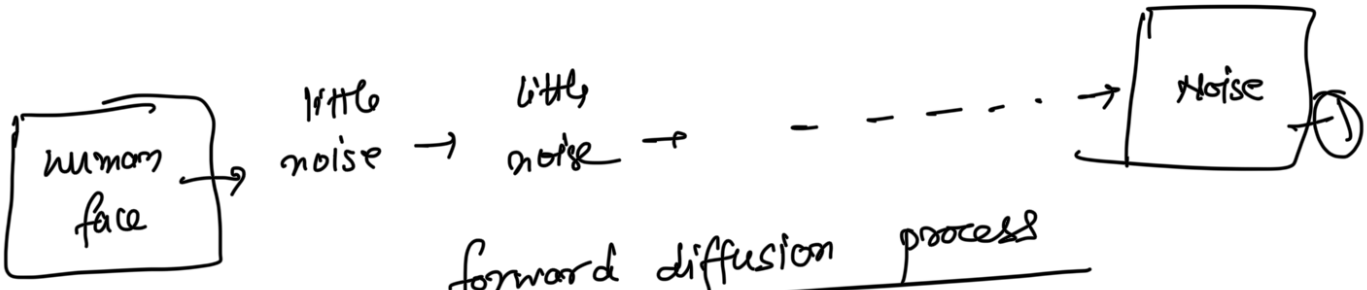
german sentence

dog → Hund

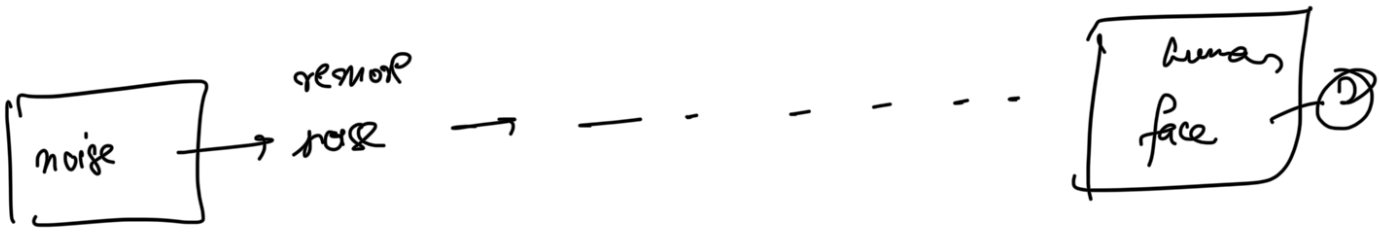
DIFFUSION MODELS

U-Net → uploaded

"Diffusion models are incremental updates where the assembly of the whole gives us the encoder decoder structure"

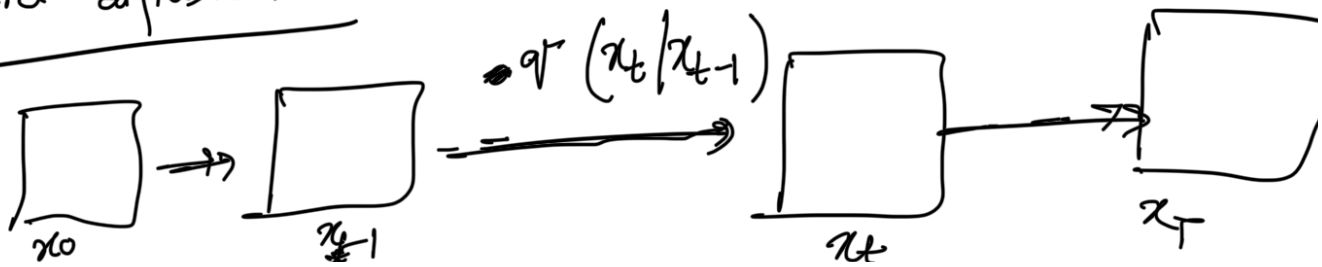


forward diffusion process



reverse diffusion

Forward diffusion



clean ← noise

zero mean unit variance

$$\boxed{x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon_{t-1}}$$

noise

std gaussian noise with 0 mean and unit variance

Note → variance of x_t → remain constant over time

x_{t-1} zero mean and unit variance

$\sqrt{1-\beta_t}\, x_{t-1}$ $\qquad$ $1-\beta_t$

$f(x) \longrightarrow$ var. A

$a f(x) \longrightarrow$ var. $a^2 A$

$\sqrt{\beta_t}\, \epsilon_{t-1}$ $\qquad$ β_t

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon_{t-1}$$

$$Var(x_t) = 1-\beta_t + \beta_t = 1$$

unit variance

$$\boxed{q(x_t \mid x_{t-1}) = \mathcal{N}(x_t ; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)}$$

The Reparameterization trick

$$x_0 \to \cdots x_{t-1} \to x_t$$

$$x_0 \to \quad \text{—} \quad x_{t-10} \to x_{t-9} \to x_{t-8} \quad \cdots$$

$$\boxed{x_{t-10} \longrightarrow x_t}$$

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon_{t-1} \quad \text{—} \quad (1)$$

$$\alpha_t = 1-\beta_t \qquad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_t$$

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, \epsilon_{t-1}$$

$$x_{t-1} = \sqrt{\alpha_{t-1}}\, x_{t-2} + \sqrt{1-\alpha_{t-1}}\, \epsilon_{t-2}$$

$$x_t = \sqrt{\alpha_t}\left\{\sqrt{\alpha_{t-1}}\, x_{t-2} + \sqrt{1-\alpha_{t-1}}\, \epsilon_{t-2}\right\} + \sqrt{1-\alpha_t}\, \epsilon_{t-1}$$

$$x_t = \sqrt{\alpha_t}\sqrt{\alpha_{t-1}}\, x_{t-2} + \sqrt{\alpha_t}\sqrt{1-\alpha_{t-1}}\, \epsilon_{t-2} + \sqrt{1-\alpha_t}\, \epsilon_{t-1}$$

$$x_t = \sqrt{\alpha_t \alpha_{t-1}}\, x_{t-2} + \boxed{\sqrt{1-\alpha_t\alpha_{t-1}}}\, \epsilon$$

$$\boxed{x_t = \text{— — . . . —}}$$

$$x_0 \leftarrow \quad \boxed{x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon} \longrightarrow$$

$\sqrt{\alpha_t \alpha_{t-1} \alpha_{t-2} \cdots \alpha_0}$

x_0

x_t

$$\boxed{q(x_t / x_0) = \mathcal{N}(x_t ; \sqrt{\bar{\alpha}_t}\, x_0 , (1-\bar{\alpha}_t) I)}$$

Diffusion Schedule :

$$q(x_t / x_0) = \mathcal{N}(x_t ; \sqrt{\bar{\alpha}_t}\, x_0 , (1-\bar{\alpha}_t) I)$$

$$\alpha_t = 1 - \beta_t \quad \text{and} \quad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$$

$\beta_t \neq$ Same

Completely different β at different steps

β_t $(\bar{\alpha}_t)$ changing with time is called the diffusion Schedule

$$\boxed{\beta_t \rightarrow 0.0001 \text{ to } 0.02}$$

Diffusion times are equally spaced

Cosine diffusion Schedule

$- - \cos^2\left(t \cdot \frac{\pi}{\ldots} \right)$

$\cos^2\theta + \sin^2\theta = 1$

$\alpha_t = \cos\left(\frac{t}{T} \cdot \frac{\pi}{2}\right)$

$$\boxed{x_t = \cos\left(\frac{t}{T} \cdot \frac{\pi}{2}\right) x_0 + \sin\left(\frac{t}{T} \cdot \frac{\pi}{2}\right) \epsilon}$$

$T = 100$ $\quad$ $t \to 1 \ldots 100$

REVERSE DIFFUSION PROCESS

$P_\theta(x_{t-1} \mid x_t)$ $\quad$ Denoising

approximating the reverse distribution

$\mathcal{N}(0, I) \longrightarrow$ New Image

$x_0 \xrightarrow{t \text{ steps}} x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$

$\bar{\alpha}_t$

$\downarrow$

$\xleftarrow{\epsilon}$ NN

DENOISING DIFFUSION MODEL

$\rightarrow$ Training

repeat

$x_0 \sim q(x_0)$

$t \sim \text{uniform}(1, \ldots, T)$

$\epsilon \sim N(0, I)$

Take gradient descent step on

$$\nabla_\theta \left\| \epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, t\right) \right\|^2$$

Until convergence

---

* U-Net denoising model → Shared U-Net paper